

AST 201

Stars & Galaxies

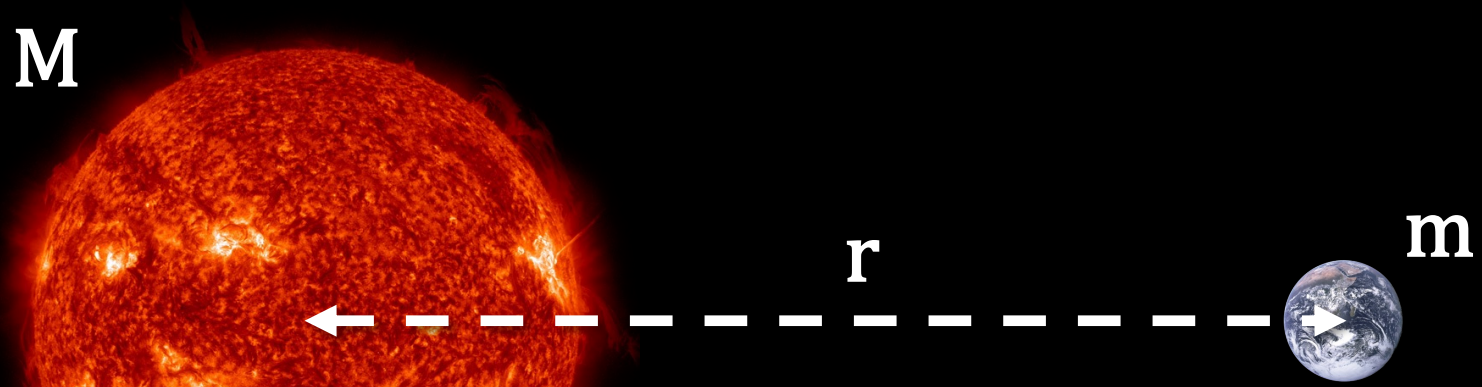
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Credit: NASA, ESA, CSA, STScI, Webb ERO Production Team

Understanding Gravity and Spacetime

Back in high school, you probably learned **Newton's law of gravity:**

The force of gravity between two objects is equal to the product of their masses and the gravitational constant, divided by the square of the distance between them.



The gravitational constant, G, which quantifies the strength of the force of gravity

$$F_{gravity} = \frac{GMm}{r^2}$$

*Nothing here is to scale

Images: NASA, SDO

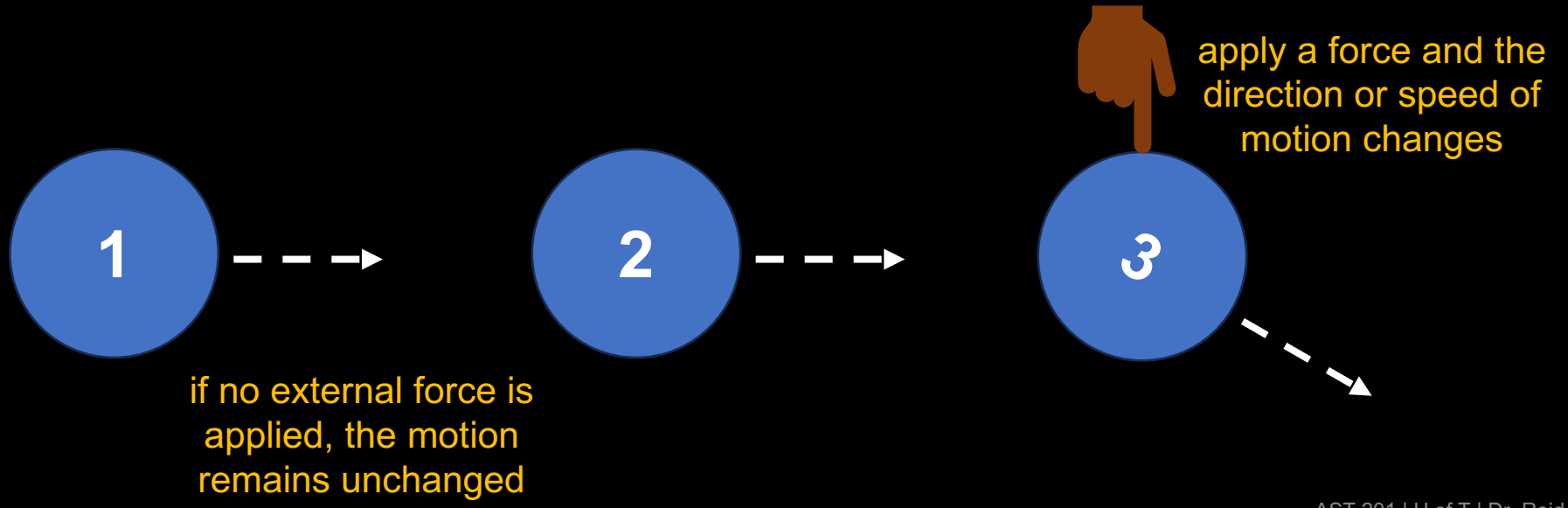
Newton's Law of gravity works well in most cases.

But what *is* this gravitational force it describes?

Let’s step back and think about how forces work. The basic rules were worked out by Newton and codified in **Newton’s Laws of Motion**.

Newton's First Law

An object in motion remains in motion with **constant speed and direction** unless acted on by an external force.



Newton's Second Law

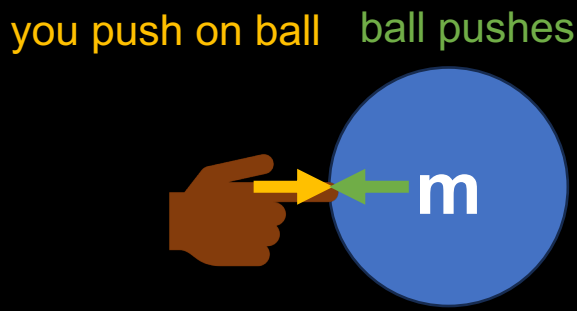
The change in speed (acceleration), a, is equal to the applied force, F, divided by the mass of the object, m.

$$a = F/m \quad \text{(or 'F = ma')}$$



Newton's Third Law

For every applied force, there is an equal and opposite reaction force.



$$F_{\text{you on ball}} = F_{\text{ball on you}}$$

Newton's Laws tell us how forces *work*.
But what *is* a force?

Modern physics describes the universe using four **fundamental forces**. One of them is gravity.

Let’s contrast gravity with the other three.

The Four Fundamental Forces of Nature

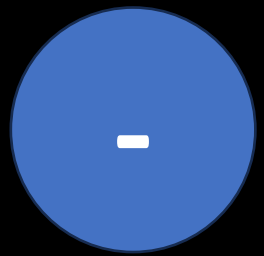
Force	"Charge"*	Strength	Range	Role
Strong nuclear force	"color" charges	1	10^{-15} m	Holds atomic nuclei together
Electromagnetism	+ and – electric charges	1/137	Infinite	Most of the forces you experience in your daily life are manifestations of this force
Weak nuclear force	weak isospin and weak hypercharge	10^{-5}	10^{-17} m	Involved in radioactive decay
Gravity	mass/energy	**10^{-39}**	Infinite	Makes masses attract one another

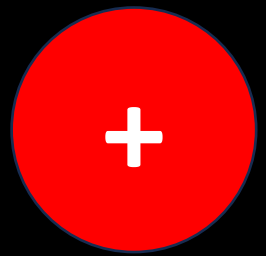
*I'm oversimplifying the concept of "charge" here, but this is a good starting place.

The fact that gravity is so much weaker than any other force is our first hint that there is something unusual about it.

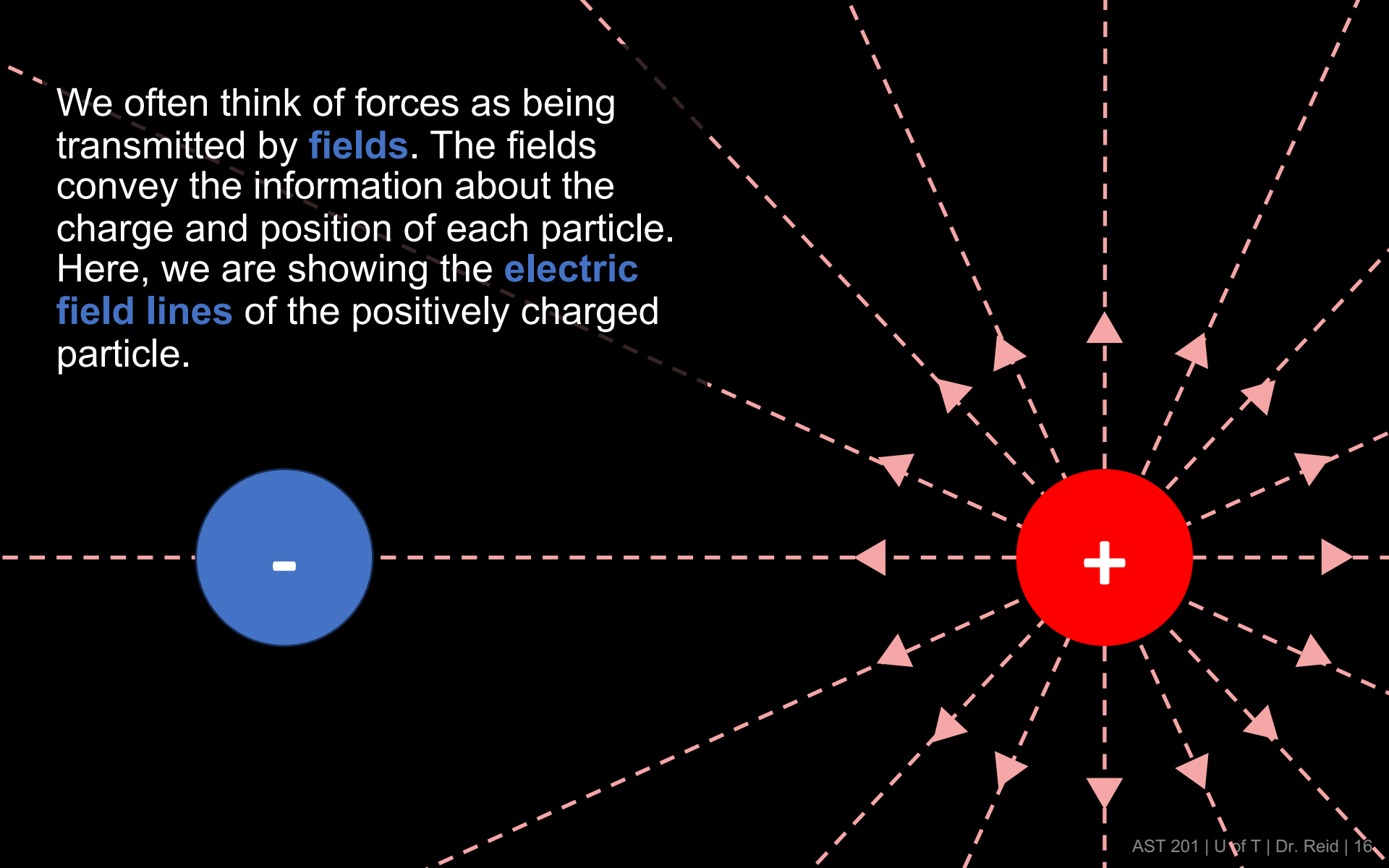
How are fundamental forces *transmitted* from one object to another?

Taking the electromagnetic force as an example, how does a negatively charged particle “know” to be attracted to a positively charged particle? If the positively charged particle moves, how does the negatively charged particle find out?



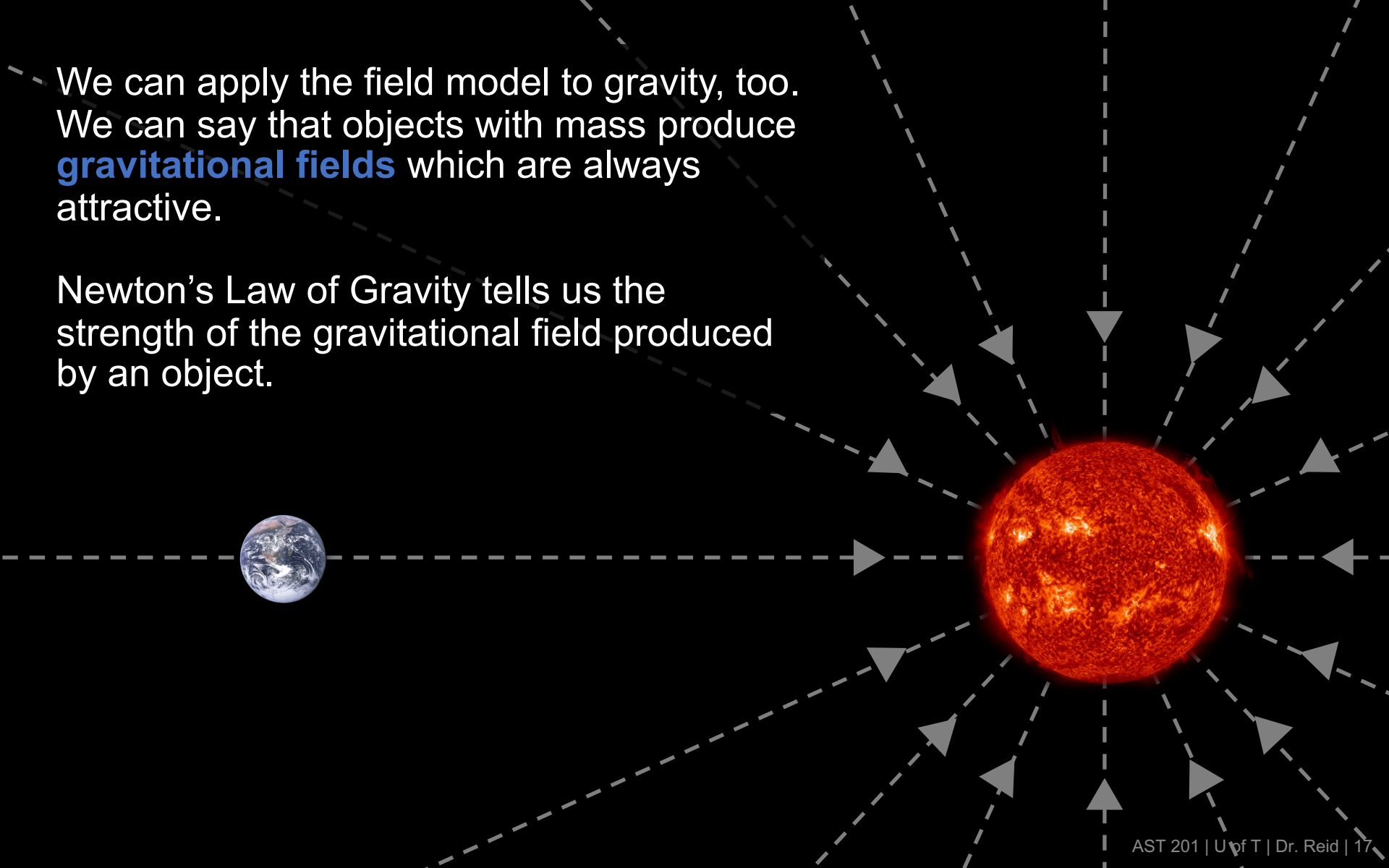


We often think of forces as being transmitted by **fields**. The fields convey the information about the charge and position of each particle. Here, we are showing the **electric field lines** of the positively charged particle.



We can apply the field model to gravity, too. We can say that objects with mass produce **gravitational fields** which are always attractive.

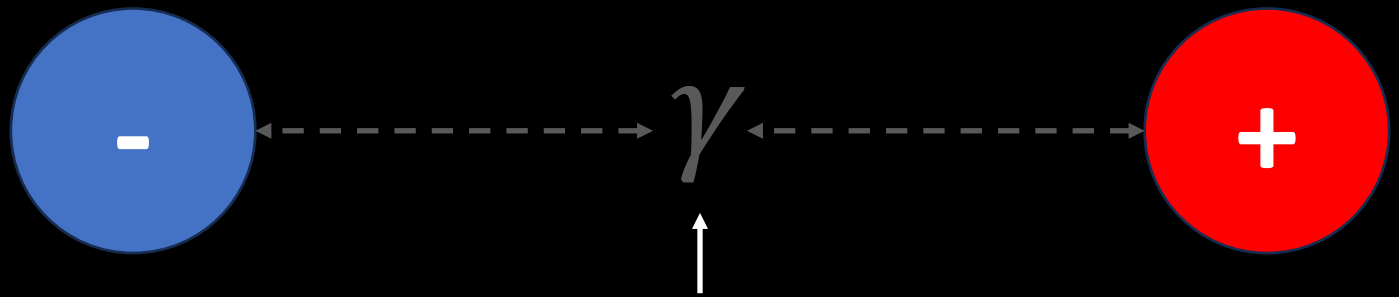
Newton's Law of Gravity tells us the strength of the gravitational field produced by an object.



In the 20th century, with the development of **quantum mechanics**, we learned that fields are “quantized”, meaning that for each field there is an associated carrier particle.

Very loosely, we can think of the fields as being “made” of these particles.

In the quantum mechanical view, forces are transmitted or “carried” by particles. Each force is carried by different particles. The carriers of the electromagnetic force are photons (γ) which are particles of light!



Electrically charged particles exchange (virtual) photons, which “carry” the electromagnetic force.

So far, we have identified and experimentally detected the force carrying particles for every force…except gravity.

Force	Carrier Particle
Strong nuclear force	gluons
Electromagnetism	photons (light)
Weak nuclear force	W and Z bosons
Gravity	**gravitons???**

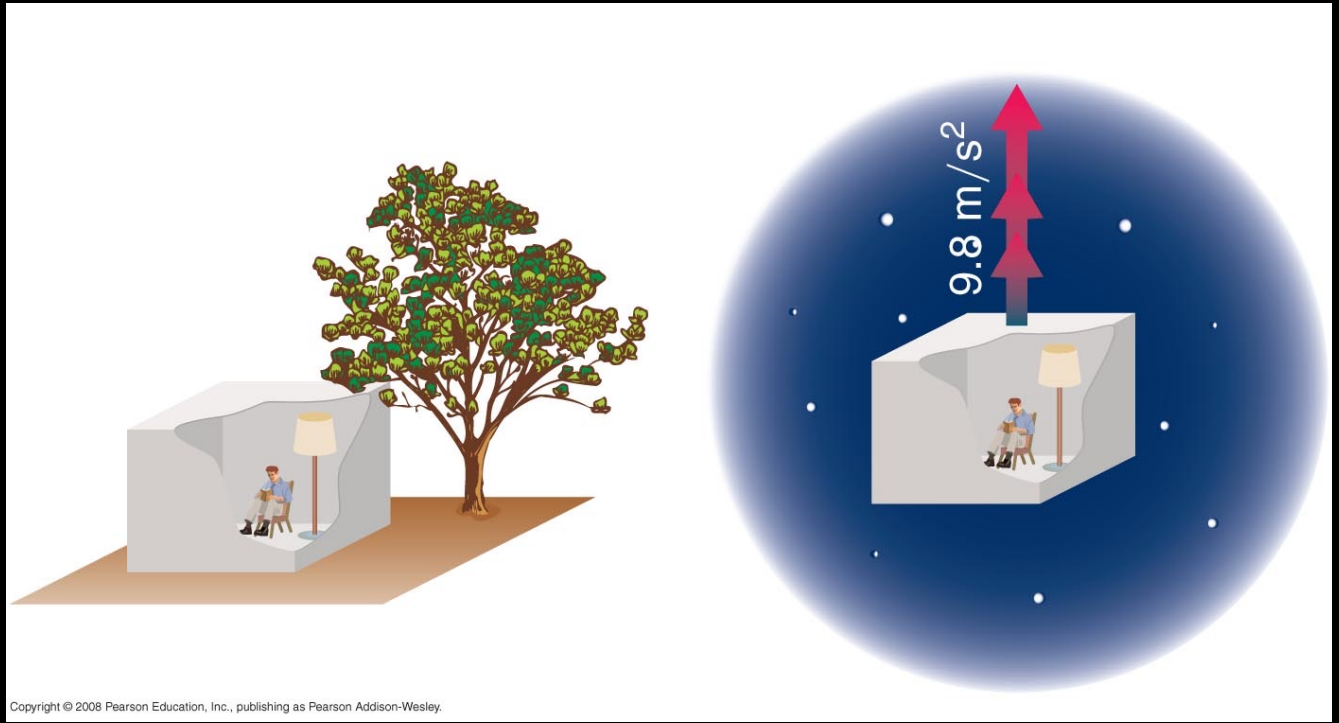
Despite about a hundred years of work, we have never found “gravitons”. So we know there’s something big we don’t understand.

The **Newtonian model of gravity**, with fields and forces, works great for most situations. But not every situation. It doesn't explain things like the orbit of the planet Mercury, or how black holes work.

Recognizing these problems, Einstein came up with a completely different way to think of gravity. Einstein's model for gravity, called **general relativity**, works in even more cases than Newtonian gravity. But still not every case...

Let’s try another thought experiment:

Imagine you wake up in an enclosed room with no windows or doors. There’s a ball on the floor. You pick it up and then let it go. It falls to the floor. Where could you be?



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Whether you are in a gravitational field with a downward acceleration of 9.8 m/s^2, or being accelerated upward at 9.8 m/s^2 in the absence of gravity, a dropped ball will still fall to the floor at the same rate.

This idea is called the **equivalence principle**.

The equivalence principle tells us that we can't distinguish acceleration from gravity.*

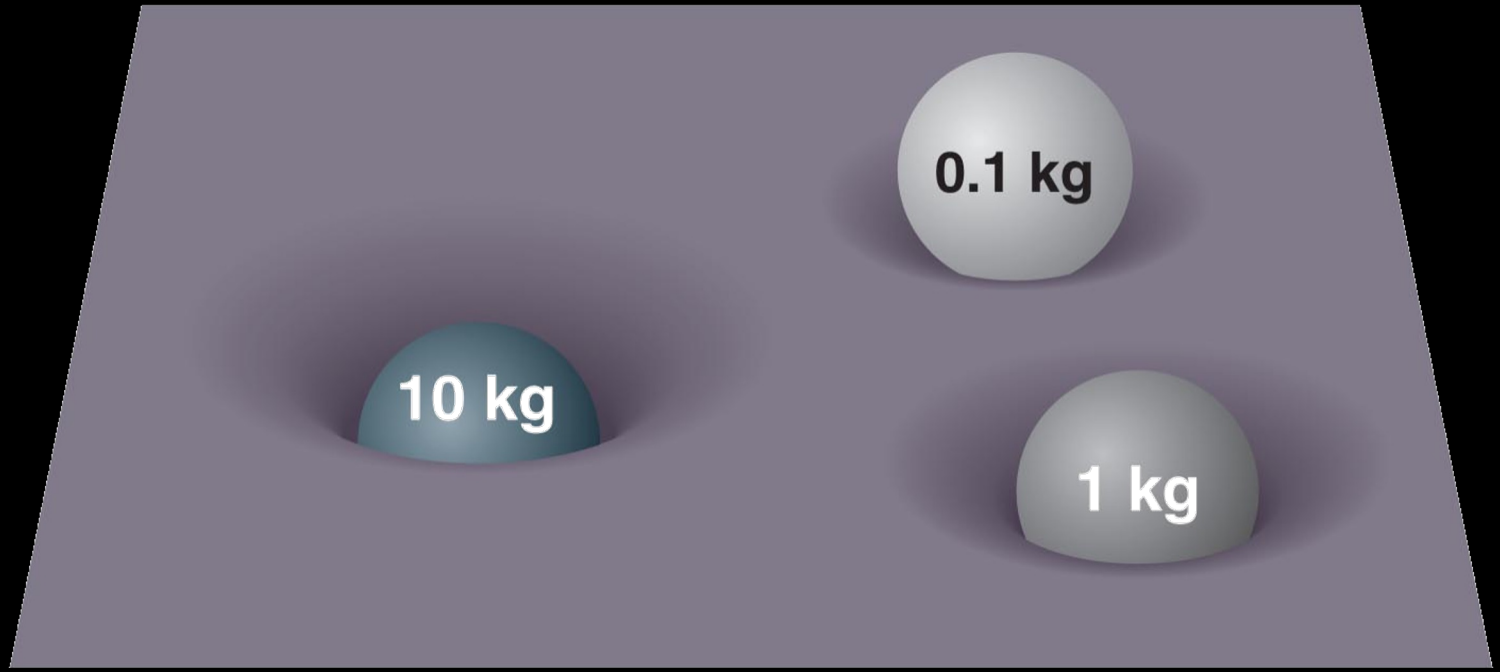
*Over very small volumes of space. Ask me why if you're curious.

General relativity, which is based on the equivalence principle, replaces Newtonian gravity with a completely different phenomenon: **curved spacetime**.

Instead of imagining gravity as a force transmitted by a force carrier particle like the other fundamental forces, Einstein *replaced* gravity with a new idea: **that objects try to follow the straightest possible path through curved spacetime**.

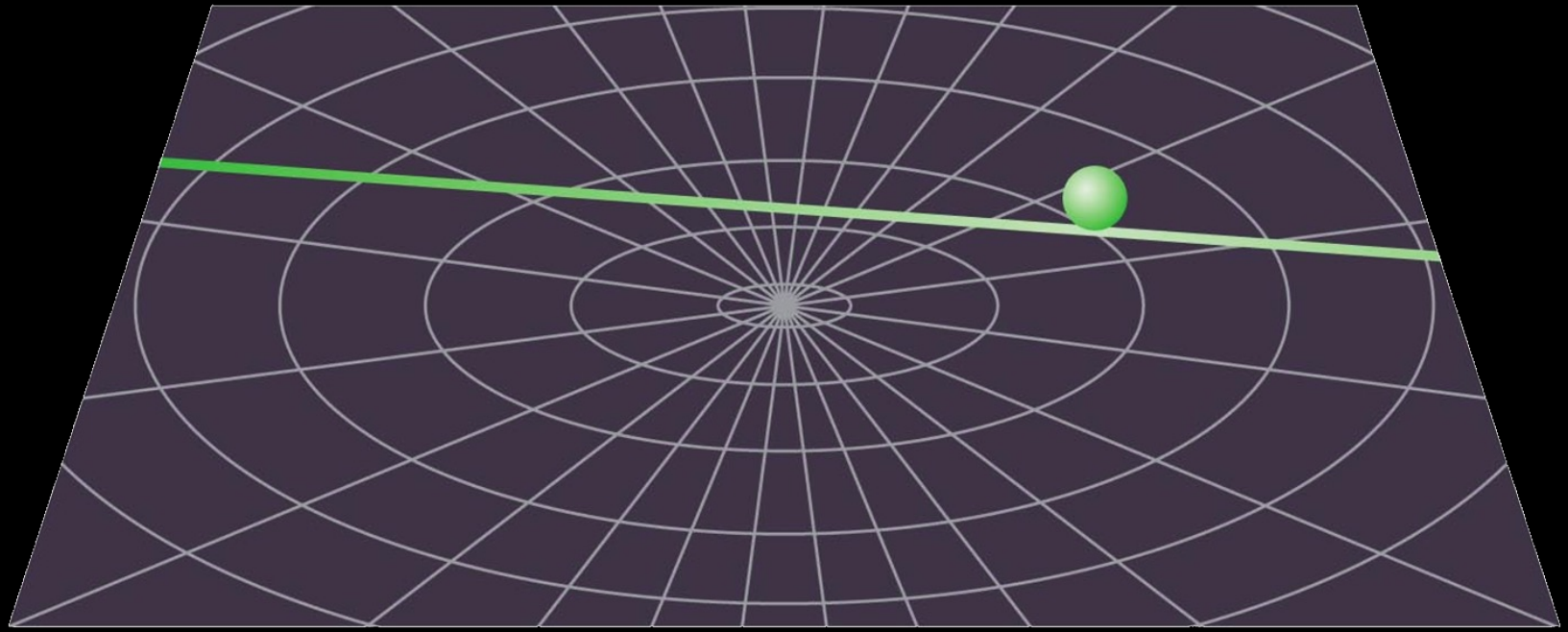
This is a totally different way of thinking of a force. In the pre-relativity way of thinking, gravity and the other fundamental forces are like the directors of a play, guiding actors around on a fixed, flat stage.

General relativity imagines gravity as warped spacetime. **Gravity becomes the stage, not one of the actors. That stage is no longer fixed and often not flat.**



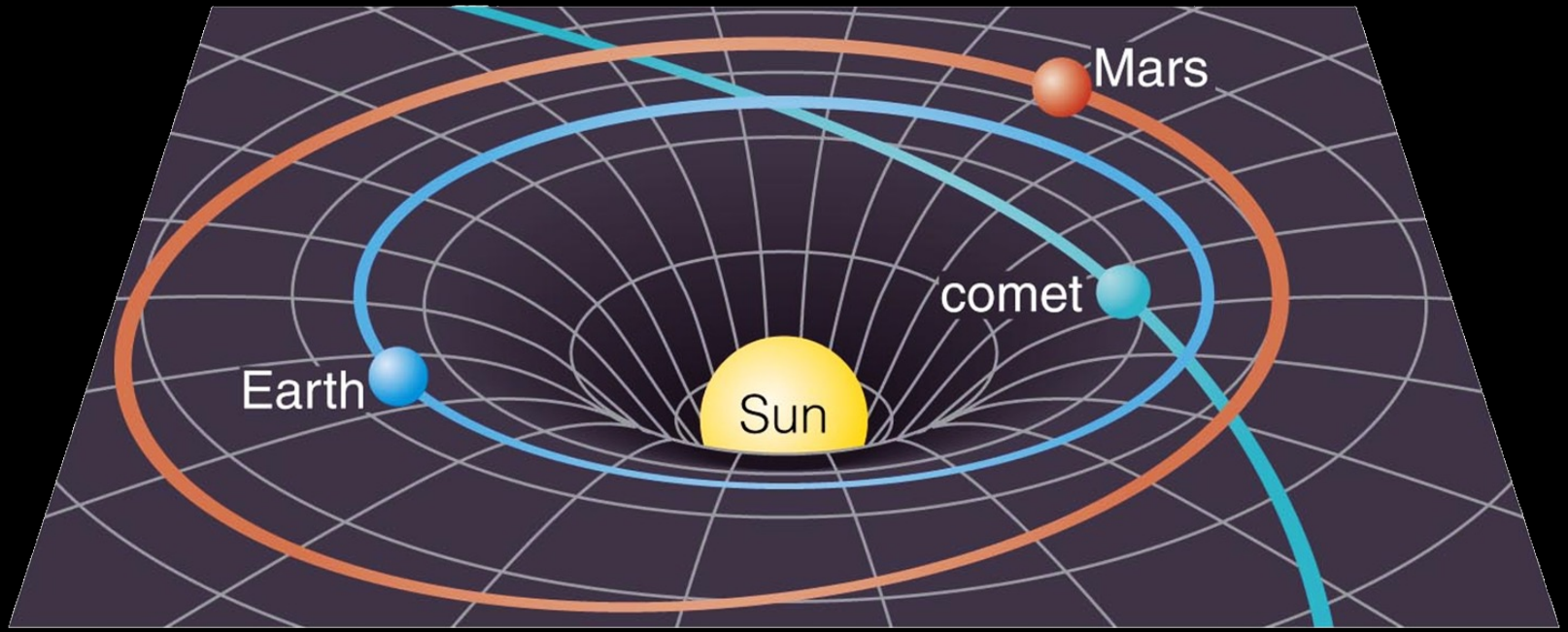
In general relativity, we can think of spacetime in analogy with a rubber sheet. Objects with mass bend the sheet.

Credit: Person Education



In general relativity, both light and matter follow the straightest possible path through curved spacetime. If the spacetime is flat, as here, they follow straight lines.

Credit: Person Education



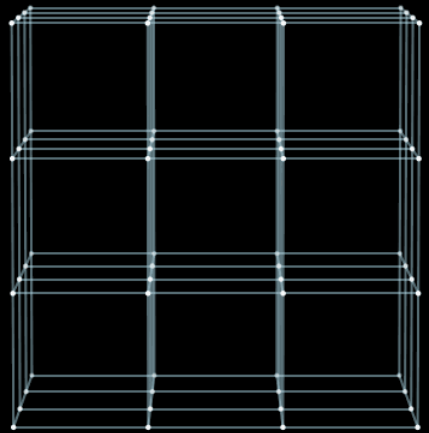
But if the spacetime is curved by the presence of a mass, each object must follow the straightest possible trajectory through that curved spacetime.

(for simplicity, this image doesn't show the curvature due to the planets or the comet.)

Credit: Person Education

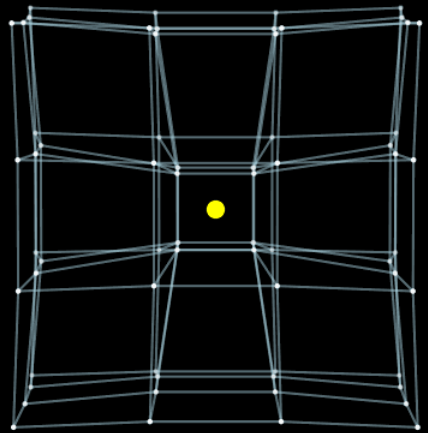
The rubber sheet analogy often confuses people, making them think that space is literally a flat sheet. We use a two-dimensional rubber sheet because it's what we can draw on the page of a book.

Let me try to show you how spacetime curvature would look in three dimensions of space.



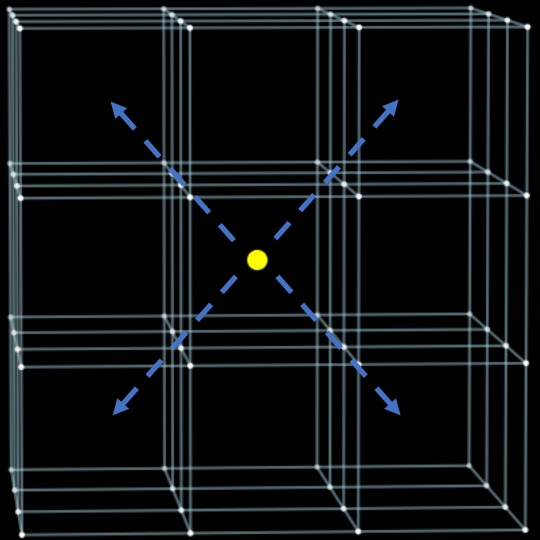
Flat spacetime

By “flat” we mean it’s not curved by the presence of mass.

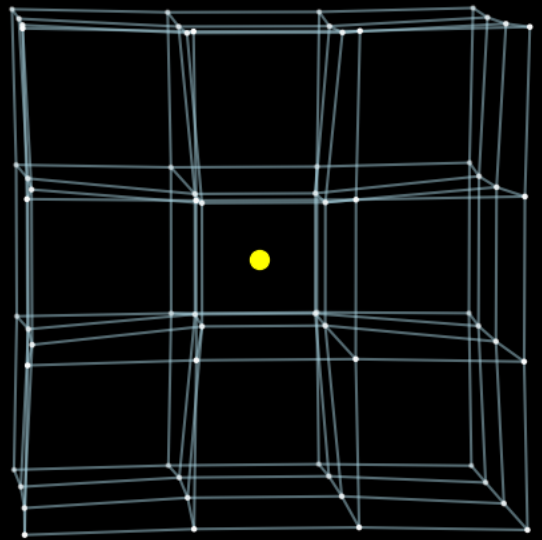


Curved spacetime

The yellow mass deforms the three-dimensional structure of space, drawing it “inward” toward the mass.



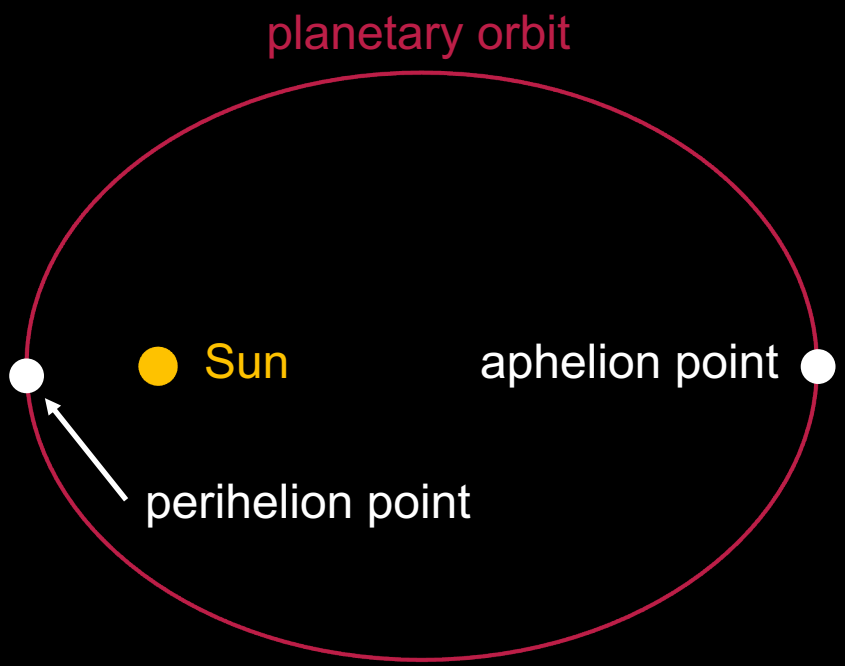
Newtonian gravity imagines gravity as a **force** transmitted through flat space.



Einstein's general relativity imagines gravity as **curvature of spacetime itself**.

Although it seems like a strange idea, general relativity has been rigorously tested since 1918 and has always passed perfectly.

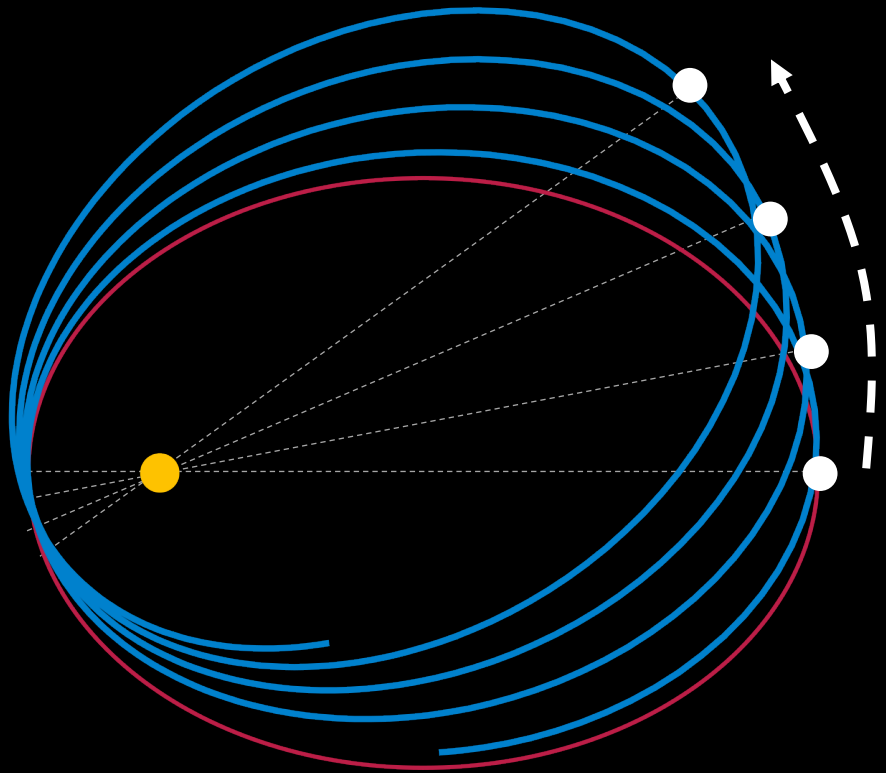
Better yet, general relativity explains features of gravity that are not explained by Newton's Law.



We usually imagine that planets orbit around stars in elliptical orbits with fixed orientations.

So, in this case, we imagine a planet going endlessly around and around the red ellipse.

Credit: KSmrq (modified), CC BY-SA 3.0



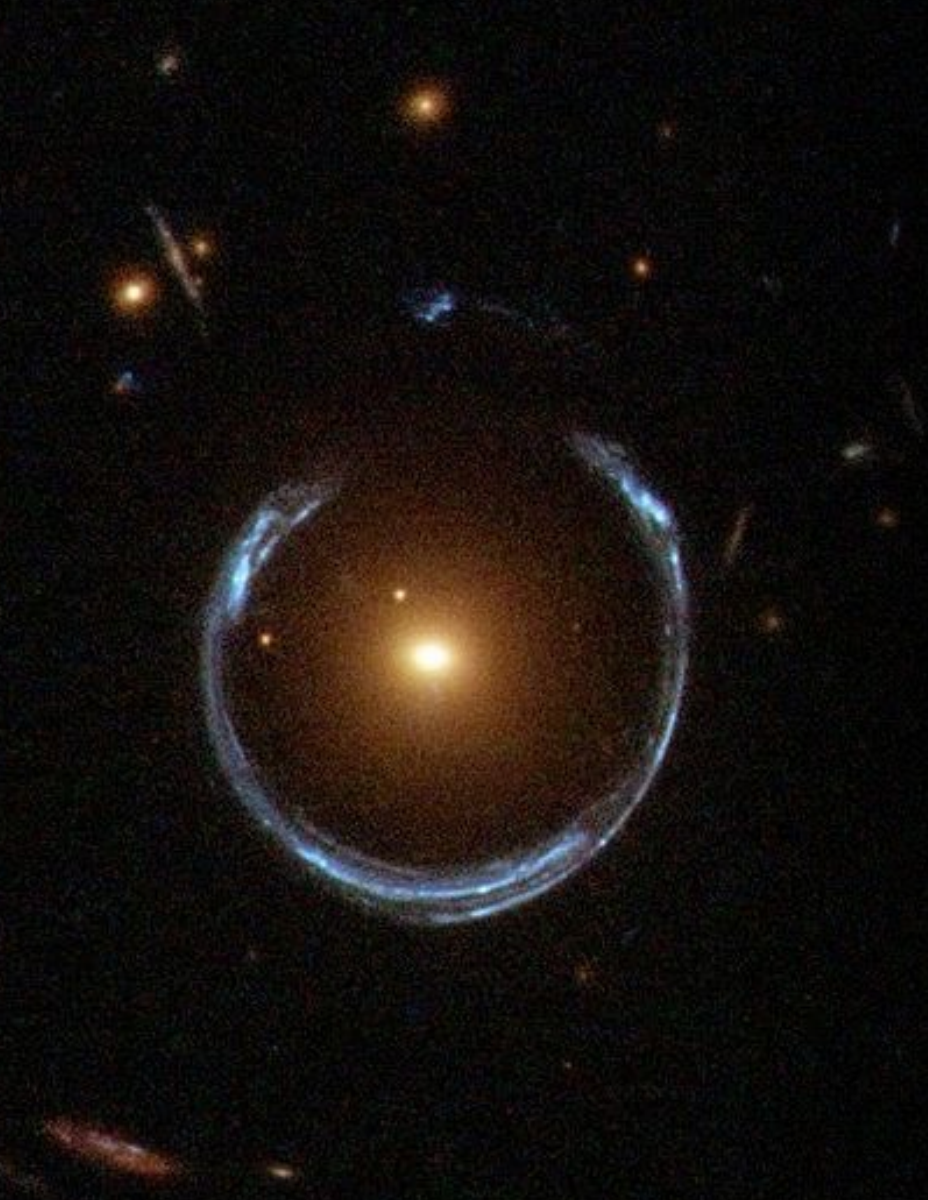
However, the long axis of a planet's orbit actually **precesses** over time. The long axis of the orbit *itself* spins around the Sun.

In our solar system, this is particularly significant in the case of Mercury, which orbits closest to the Sun.

Newton's law of gravity is not able to correctly explain this **precession of the perihelion of Mercury**. General relativity explains it perfectly.

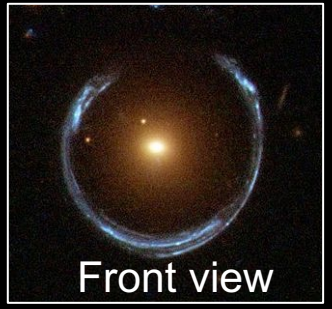
(It's actually easier to see the procession of the aphelion point, so that's what I've drawn, but note that the perihelion point is precessing too.)

Credit: KSmrq (modified), CC BY-SA 3.0



General relativity also correctly describes **gravitational lensing.**

It explains and correctly quantifies how massive objects bend the space around them forcing light to follow its curved contours.



Side view: light from the background galaxy is lensed by the spacetime curvature of the foreground galaxy, permitting us to see things we would not normally be able to see.

background blue galaxy

massive foreground galaxy bends spacetime

We summarize general relativity by saying **“matter and energy tell spacetime how to curve, curved spacetime tells matter and energy how to move.”**

Don’t forget the “time” part of spacetime. Matter and energy bend space and time.

This gives rise to the phenomenon of **gravitational time dilation:** time runs slower in places of strongly curved spacetime relative to flat spacetime.

We will return to this idea when we discuss black holes. If you want a preview, watch the movie *Interstellar*.



Several of you have mentioned the film Interstellar, which includes a beautiful and generally accurate depiction of people experiencing gravitational time dilation near a black hole.